

The Cost of Not Being Seen:

Automated Licence-Plate Reader Exposure on American Commutes and the Price of Avoiding It

TRAMES Project

kara@soulstone.org

September 22, 2026

Abstract

Automated licence-plate readers (ALPR) have moved from tollbooths to residential intersections, and a single vendor—Flock Safety—now accounts for 80.0 % of the 142,991 installations mapped in North America. Public argument about this technology is conducted almost entirely in the abstract. We supply a measurement instead: how much ALPR surveillance a typical American commuter actually drives through, and what it would cost them to refuse it.

We build a continental routing engine that models ALPR fields of view as directional geometry—98.0 % of mapped cameras carry a bearing—and can be asked for the fastest route that crosses none of them. Routing 17,580 real home–work commutes drawn from Census LEHD LODES data across 15 states, twice each, yields both quantities on identical road network.

Three findings. First, 78.7 % of commutes pass at least one ALPR, a median of three and a 99th percentile of thirty-five. Second, refusal is far cheaper than expected: 82.1 % of commutes can be routed to *zero* ALPR exposure for a median of 1.98 min, an overhead of 7.90 %. Third, and most consequentially for policy, demographic gradients that appear robust nationally—origin tracts in the highest quartile of Black population pass 1.36× as many cameras as the lowest, with disjoint confidence intervals—vanish when tracts are ranked within their own county (1.05×, indistinguishable). Mapped camera density varies 3.4× per capita across the states we study. We therefore report the equity result as a between-county phenomenon that this design cannot attribute to siting, mapping effort, or urban form, and we argue that no study built on crowdsourced surveillance data should report such a gradient without this check.

Resolving exposure by vendor locates the disparity: tolling and transport-agency cameras show a gradient of 1.62× against 1.28× for Flock, whose installations are four-fifths of the network. Per unit deployed, non-Flock cameras reach 2.3× as many commuters. Flock’s gradient dissolves under within-county ranking; the transport-agency gradient narrows to 1.20× and remains distinguishable—the one contrast in the paper that survives the control. Routing the same commutes against a second snapshot taken two months later, with 18 % more mapped cameras, raised measured exposure in exact proportion and left the marginal price of evading a camera unchanged.

All code, data acquisition, and analysis are reproducible from public sources.

1 Introduction

Automated licence-plate readers photograph every passing vehicle, resolve the plate, and retain the read. Unlike a traffic camera they are not triggered by an offence, and unlike a patrol officer they do not tire. Fixed, pole-mounted units have spread from limited-access highways into residential neighbourhoods, arterial roads, and the entrances of private subdivisions.

The public debate about this expansion is conducted in the vocabulary of policy: retention windows, warrant requirements, audit logs, data-sharing agreements. What that vocabulary lacks is a measurement of the thing being regulated. How much of this does a person actually drive through on an ordinary Tuesday? And if they would rather not, is declining a practical option or a theoretical one?

This paper answers both, and does so by building the instrument the question implies. We construct a routing engine that treats ALPR fields of view as first-class geometry and can be asked for the fastest route that avoids them. Running each sampled commute through it twice—once normally, once avoiding—yields the exposure and the price of refusing it, for the same trip, on the same network, under the same traffic model.

The second quantity is where the paper earns its keep. A privacy protection that costs four minutes is a feature. One that costs forty is available only to people with forty minutes to spare, and a right that is priced is not evenly held. We therefore report the cost of refusal across origin-tract income and racial composition—subject to the substantial caveat developed in Section 7.2 and quantified in Section 5.5, which turns out to govern how the demographic results may be read.

1.1 Contributions

- **A directional exposure model at continental scale.** Cameras are wedges, not circles. 98.0 % of mapped North American ALPR installations carry a bearing tag, making this feasible nationally. Proximity-based counting inflates exposure by charging drivers for cameras aimed at the opposite carriageway; we do not do it.
- **Avoidance routing with no forked router.** 135,210 camera cones are compiled into a stock GraphHopper spatial index at import time, so an ordinary per-request custom model can reference them by name. No engine patches, no geometry in the request.
- **A commute-weighted rather than area-weighted sample.** 17,580 real home–work pairs drawn from LEHD LODES with probability proportional to worker count.
- **A vendor-resolved account of exposure.** Flock and the DOT/tolling vendors follow different deployment logics—who *buys* surveillance versus where road infrastructure sits. Separating them shows the national racial gradient is substantially stronger in transport infrastructure than in the vendor that dominates both the network and the public argument about it.
- **A coverage-bias control that changes the conclusion.** We show that the national demographic gradient does not survive within-county ranking, quantify the 3.4× per-capita spread in mapping intensity that plausibly drives it, and argue this check should be mandatory for work of this kind.
- **A paired re-measurement.** The same commutes routed against a snapshot taken two months later with 18 % more mapped cameras: exposure rose in proportion, the marginal price of evading a camera did not move, and the states we had flagged as under-mapped are the ones that grew most.

2 System

2.1 Why a fork, and of what

The measurement needs a router that treats “avoid these 135,210 polygons” as a cost rather than a post-hoc filter, and the eventual application needs to be a real navigation client rather than a

research script.

OsmAnd (GPLv3) is the base: the only mature Android navigation client with first-class OpenStreetMap support, offline maps, and a documented online-routing extension point. Its *code* is GPLv3; its *artwork* is CC-BY-NC-ND 4.0 and must therefore be replaced rather than modified in a derivative.

GraphHopper 11.0 is the routing engine. Valhalla was evaluated and rejected: its `exclude_polygons` support carries open correctness defects (#3266, #4659, #5069) whose failure mode is silently routing *through* the excluded region—which is precisely the measurement everything here depends on.

One architectural property made the rest tractable. OsmAnd’s online routing engines return *geometry only*; turn instructions are re-derived client-side from offline map data. A custom online router therefore never has to synthesise navigation instructions.

2.2 The keystone: import-time custom areas

GraphHopper resolves `custom_areas.directory` into a spatial index *at graph import time*. We verified empirically—the documentation is ambiguous—that an area registered this way can be referenced by name from a *per-request* custom model:

```
{ "priority": [ { "if": "in_alpr", "multiply_by": 0.01 } ] }
```

This is what makes the design possible without forking the engine; shipping 135,210 polygons in every routing request is not viable. The cost is that regenerating cones requires a full re-import (82 min to 113 min for North America, by load).

Two further configuration decisions are load-bearing:

- **Contraction Hierarchies are disabled.** CH bakes edge weights into the prepared graph, which is incompatible with a per-request avoidance strength. Landmarks (LM) are used instead.
- **The `car_alpr` profile shares car’s LM preparation.** This halves preparation time and is valid *only* because avoidance multiplies priority strictly downward, so the borrowing profile’s weights never exceed the preparation profile’s on any edge—the condition LM correctness requires.

2.3 Deployment posture

The routing server binds to localhost. It has no authentication and no rate limiting and is not internet-facing. Any public deployment must sit behind a reverse proxy supplying both.

3 Data

All sources are public and key-free.

3.1 ALPR installations

Source: OpenStreetMap via the Overpass API, selecting `man_made=surveillance` with `surveillance:type=ALPR`, matched case-insensitively (19 readers are tagged `alpr`)—the corpus surfaced by the DeFlock project, taken from the upstream rather than a mirror. Retrieved in five regional queries (continental US, Alaska, Hawaii, Canada, Mexico) and deduplicated by OSM node id. The results in this paper rest on the snapshot of **2026-09-22: 142,991 unique camera nodes** (Figure 1). A first snapshot, taken 2026-07-24, held 120,838; Section 5.7 routes the same commutes against both.

Table 1: Directional and vendor tagging across the two snapshots. The direction-key rows overlap (392 and 415 nodes carry both keys). Vendors are matched as case-insensitive substrings under any of the four vendor keys, in both columns.

Property	2026-07-24		2026-09-22	
	Count	Share	Count	Share
Camera nodes	120,838		142,991	
Any direction tag	119,249	98.7 %	140,112	98.0 %
via direction	117,356	97.1 %	138,216	96.7 %
via camera:direction	2,285	1.9 %	2,311	1.6 %
Multi-head (;-separated)	7,014	5.8 %	9,150	6.4 %
Arc-range form (199–269)	4,806 tokens	—	4,632 tokens	—
operator tagged	22,354	18.5 %	23,872	16.7 %
Flock Safety	99,650	82.5 %	114,360	80.0 %
Motorola Solutions	5,993	5.0 %	7,411	5.2 %
Genetec	2,876	2.4 %	3,721	2.6 %
Leonardo	1,020	0.8 %	1,215	0.8 %
Axis Communications	1,052	0.9 %	3,327	2.3 %
No vendor tag	5,811	4.8 %	6,850	4.8 %

A fetch that fails over between Overpass mirrors has to check the mirror’s database date. On the first attempt at the September snapshot one regional query fell through to a mirror whose database was dated 2026-05-06; that box overlaps the southern United States, regions were merged last-wins, and roughly 13,600 cameras across Texas, Arizona and Florida silently reverted to four-month-old tags while some 400 nodes deleted since May returned as ghosts. Nothing failed. The fetch now rejects any answer older than 48 hours and merges overlapping regions newest-database-first; every tile of the snapshot used here falls within a 13-minute window.

Three tagging realities shaped the parser, each contradicting a plausible assumption:

1. **direction, not camera:direction.** The OSM wiki documents `camera:direction` for surveillance nodes. Mappers overwhelmingly use plain `direction`, by a factor of 60. Reading only the documented key discards 97 % of the bearings.
2. **Arc ranges are real, and mostly 45°.** 4,632 values give an explicit range; 3,806 of them (82 %) span exactly 45°. That is what justifies 45° as the default span for the ≈ 97 % of cameras giving a bare bearing—the default is inherited from the data, not assumed.
3. **Multi-head units exist** (320;190, 0;72;144;216;288, 199–269;325–35); 9,150 nodes. These typically watch *both* carriageways, so silently taking the first bearing is the worst available error: it makes a camera that sees everyone look like one that sees half. The parser emits one wedge per head.

3.2 Cone construction

Each camera head becomes a circular sector: apex at the node, bisector along the tagged bearing, span 45° (or the tagged arc), radius 60 m. The radius must reach across the carriageway from a set-back pole; it was chosen on that reasoning and then swept over {30, 45, 60, 90} m (Section 7.3). Measured exposure is only mildly sensitive to it, but the avoidance result is not robust above the assumed value.

(a) 142,991 mapped ALPR installations, by vendor

(b) Installation density (log scale)

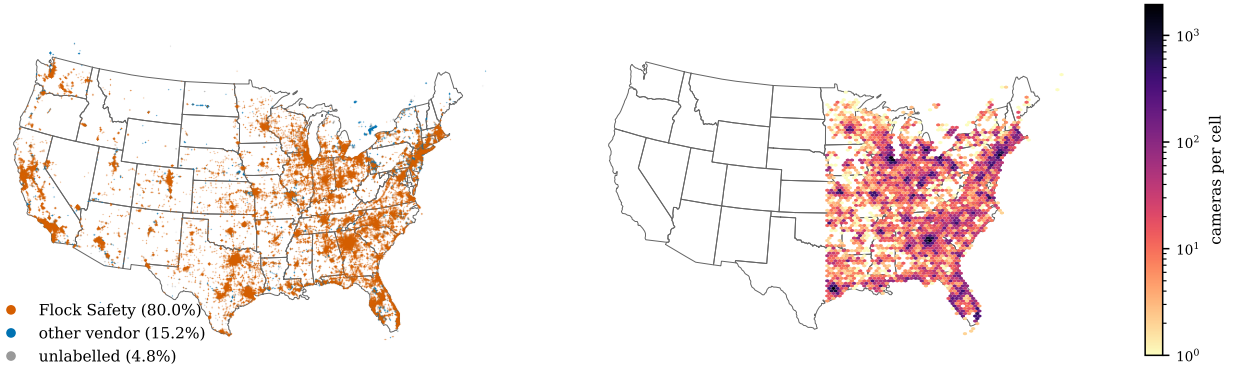


Figure 1: Mapped ALPR installations in the contiguous United States. (a) By vendor: Flock Safety dominates by count almost everywhere, with other vendors concentrated on limited-access corridors. (b) Density on a log scale. The visible structure is partly deployment and partly *mapping*: see Section 5.5.

Table 2: Vendor-split cone sets, 2026-09-22 snapshot. The cones-per-camera column is itself a result: non-Flock installations are multi-head more than six times as often (22.8 % against 3.6 %), consistent with tolling and DOT gantries covering every lane while a Flock unit watches one approach.

Set	Cameras	Cones	Cones/camera	Union parts
Flock Safety	113,611	120,366	1.06	109,427
Other named vendor	20,435	30,128	1.47	21,587
No vendor tag	5,997	6,590	1.10	5,211

All wedges are unioned into a single MultiPolygon of **135,210 parts**, published as one feature with id `alpr`. Overlapping wedges merging is correct and intended: the question is *is this road segment observed*, not *by how many devices*.

Assigning vendor requires care. Vendor appears under four keys (`manufacturer`, `surveillance:manufacturer`, `brand`, `surveillance:brand`) and is spelled inconsistently. An exact match on `manufacturer = "Flock Safety"` finds 109,892 cameras; a case-insensitive substring search across all four finds 114,360. The 4,468 difference is mostly 4,711 nodes carrying the vendor under `brand`, plus variants (“Flock Group Inc.”, “FlockSafety”). Misfiling 3.9 % of Flock cameras as non-Flock is the one error a vendor comparison cannot absorb.

Untagged cameras are held out of *both* vendor sets rather than assigned to the complement of Flock, since an untagged camera is likelier to be an unlabelled Flock unit than a non-Flock one. They are reported as their own set so nothing is silently dropped, with the caveat that their direction coverage is lower (88.0 % against 99.4 % for Flock and 93.8 % for other named vendors—the last down from 98.5 % in July, as newly mapped Axis units often arrive without a bearing).

3.3 Road network

Geofabrik North America extract, 18 GB, retrieved 2026-07-24 and held fixed for both snapshots, so that Section 5.7 isolates the camera data: every unavaoided route is identical between the two runs to the metre. Imported with `ignored_highways = footway,construction,cycleway,path,steps`. `track` is deliberately retained: unpaved tracks are plausible avoidance routes, and excluding them

would bias the cost estimate downward by removing options a determined driver has.

3.4 Commute sample

Source: LEHD **LODES8**, `od_main_JT00`, 2022—all jobs, both ends in-state, built from unemployment-insurance wage records linked to employer establishments. These are administrative counts of real jobs, not survey estimates.

Sampling design carries more weight here than sample size. Uniform random points would over-represent empty rural space where nobody commutes and no cameras exist, understating exposure badly. Instead each origin–destination pair is drawn **with probability proportional to the number of workers actually making that trip**, so the distribution approximates what an average *commuter* experiences rather than what an average *square kilometre* contains.

Block-level LODES geocodes are aggregated to **tract** (first 11 digits)—the finest resolution at which ACS demographics are reliable, and finer than the routing’s effective precision. Same-tract commutes are dropped: origin and destination centroids coincide, so there is no route to measure. Georgia alone yields 1,199,250 distinct tract pairs covering 4,190,313 workers.

1,200 pairs are requested per state; duplicate draws are collapsed, costing 0.53%, for **17,904 sampled commutes** across 15 states (AZ, CA, CO, FL, GA, IL, MO, NC, NY, OH, PA, TN, TX, VA, WA). Seed 20260725; the sample is reproducible. The sampled pairs represent 600,899 workers; the median pair carries 10 and the largest 2,832.

Because draws are taken with replacement and then deduplicated, each pair appears once and every row is weighted equally downstream. That is valid only while inclusion probability remains proportional to worker count, which decays once a pair is likely to be drawn more than once. Measured on Georgia, the largest state sample, the highest-flow pair has an expected 0.22 draws per 1,200 and a median pair 0.003, 6: the design is probability-proportional-to-size with distortion confined to the extreme upper tail (roughly 11% under-representation for the single largest pair, negligible below it). The **workers** and **flow_share** columns are carried through the pipeline but deliberately unused—equal weighting already yields the worker-weighted distribution, and weighting again would apply the flow twice.

3.5 Demographics and geography

Tract centroids are 2024 Census Gazetteer internal points (`INTPTLAT/INTPTLONG`), which are guaranteed to fall inside the tract polygon. Income is ACS 5-year 2022 table **B19013**; race and ethnicity are table **B03002**, chosen over B02001 because it is Hispanic-aware, so “non-Hispanic Black” is directly available rather than reconstructed. Demographics attach to the *origin* (home) tract: the claim under test concerns who lives where surveillance is dense.

4 Method

4.1 Two routes per commute

Each sampled pair is routed twice against the same graph: a **baseline** with no custom model, and an **avoided** route applying priority: `[{if: "in_alpr", multiply_by: 0.01}]`. Both disable CH, which per-request custom models require.

0.01 is the *maximum* avoidance strength, not a typical setting. Results therefore describe an **upper bound on both the benefit and the cost** of avoidance—the strongest available

preference short of prohibition. This is deliberate: it establishes whether meaningful avoidance is possible at all, and what it costs at the limit.

4.2 Exposure is directional, not proximate

A camera counts as having seen the driver *only if the route line intersects its cone*. Passing 20 m from a camera aimed the other way is not exposure. Implementation is an STRtree over the 135,210 cone parts for candidate lookup, then an exact intersection test.

Two conservatisms remain, both understating exposure: route geometry is a polyline while a vehicle has width, and a cone occluded by intervening buildings is still counted as full sight-line.

4.3 Vendor attribution costs no additional routing

Routing does not change between vendor questions—the avoidance model targets the all-vendor union either way—so Section 5.6 is entirely a matter of re-scoring existing routes against different cone sets. This is possible only because the experiment persists route polylines to a sidecar; keeping only counts would have made both the vendor split and the radius sweep of Section 7.3 fresh multi-hour re-runs.

As a consistency check, the three vendor sets should sum to slightly *more* than the all-vendor union, because cones from different vendors that overlap merge into one part in the union and are counted once while the split sets count them separately. Measured excess is 1.0 %: right sign, plausible magnitude.

4.4 Statistical reporting

- **Medians lead, means accompany.** Both distributions are heavily right-skewed; a minority of long commutes through dense corridors pull the mean well above the typical experience.
- **Bootstrap confidence intervals** (percentile, 2,000 resamples, seed 20260725). Camera counts are discrete and zero-inflated, nowhere near normal.
- **Quartile contrasts, not regressions.** With tract-level aggregates a regression coefficient invites a causal reading the design cannot support.
- **Total exposure and per-kilometre density are reported as peers.** A raw count gap that vanishes once normalised per km is about distance, not siting. But the converse is a finding, not a nuisance: a group whose commutes are *shorter but more densely watched* shows a rate gap and no count gap, and treating the rate purely as a control would discard it. Cameras per km is a ratio of sums bootstrapped over commutes; a mean of per-commute rates would let very short commutes dominate.
- **Ecological inference is not attempted.** Tract composition is a property of tracts, not of the individuals routed.

4.5 Execution

Twelve concurrent workers against the local routing server. Worker count was measured, not assumed: twelve completed 300 commutes in 324 s, while thirty-two took longer than 576 s on the identical set—past about twelve, contention outweighs added concurrency. Routing dominates cost overwhelmingly; the geometric exposure test is 0.2 % of per-commute work.

Failed routes are dropped rather than retried: failures are dominated by unroutable centroid pairs (islands, centroids landing off-network), properties of the sample rather than transient faults.

Table 3: Exposure and the cost of avoidance over 17,580 commute-weighted origin–destination pairs in 15 states. Median commute 24.4 km / 24.1 min.

	Median [95 % CI]	Mean
<i>Unavoided route</i>		
Cameras passed	3 [3–3]	5.63
Commutes passing ≥ 1		78.7 %
Commutes passing ≥ 5		39.7 %
Commutes passing ≥ 10		18.3 %
90 th / 99 th percentile	14 / 35 cameras	
Exposure rate	0.105 cameras/km	
<i>ALPR-avoiding route</i>		
Cameras passed	0 [0–0]	0.30
Reduced to zero		82.1 %
Extra time (min)	1.98 [1.89–2.07]	5.97
Extra distance (km)	0.76 [0.72–0.79]	2.69
Avoidance overhead	7.90 % [7.60–8.17]	14.22 %
Minutes per camera evaded	0.78 [0.77–0.80]	1.31

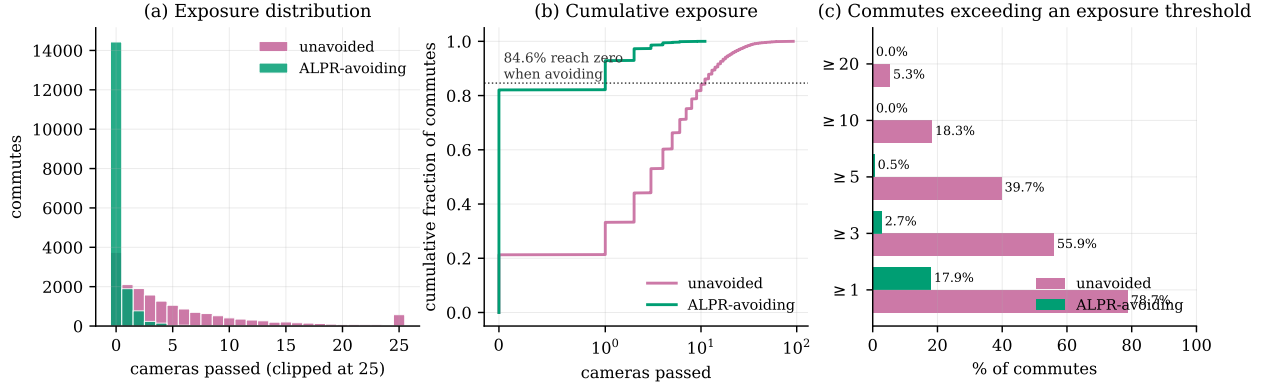


Figure 2: Exposure before and after ALPR-avoiding routing. (a) Distributions; (b) cumulative, log-scaled abscissa; (c) share of commutes exceeding a threshold. The zero-exposure mass moves from 21.3 % to 82.1 %.

17,580 of 17,904 commutes routed successfully (1.8 % dropped)—the same 324 failures in both snapshots. The September run took 10.6 h against 5.4 h for July on the same hardware: with a fifth more penalised edges, the avoiding search visits more of the graph.

Commutes are processed in seeded random order. The sample is grouped by state and the executor preserves input order, so in file order results arrive one state at a time: an interruption would leave some states complete and others absent, and no partial output could support a national estimate. Shuffling makes any prefix a near-uniform sample of the whole. This proved material rather than hypothetical—an earlier single-state interim read gave a median avoidance overhead of 22.7 % against that run’s national 5.96 %.

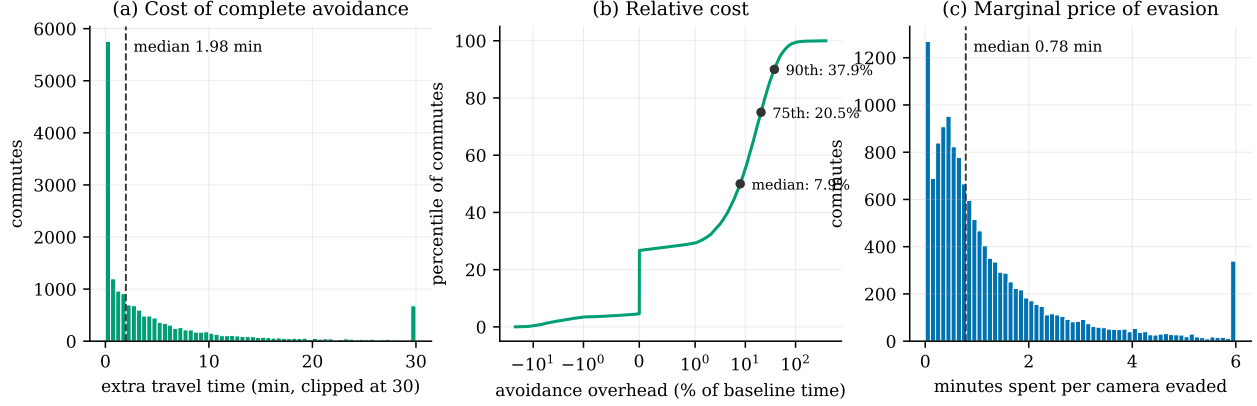


Figure 3: The price of complete avoidance. (a) Absolute extra time; (b) overhead as a percentage of baseline, by percentile; (c) marginal minutes per camera evaded. All three are heavily right-skewed—a small tail of commutes pays a great deal, while the typical commute pays almost nothing.

5 Results

5.1 Exposure on the unavoided commute

78.7 % of American commutes pass at least one ALPR. The median commuter is read three times each way; the 90th percentile fourteen times and the 99th thirty-five. Aggregate exposure is 0.105 cameras per kilometre driven.

5.2 The cost of refusal

82.1 % of commutes can be routed to zero ALPR exposure, at a median cost of 1.98 min and 0.76 km—an overhead of 7.90 %. The median marginal price is 0.78 min per camera evaded.

For four American commuters in five, then, complete avoidance of licence-plate surveillance costs about two minutes. This figure is conditional on the assumed camera field of view: it is stable if the true radius is smaller than the 60 m assumed here and degrades if it is substantially larger, which Section 7.3 quantifies. This is the paper’s central quantitative claim, and it is considerably more favourable than the metropolitan spot-checks that motivated the study suggested.

The distribution matters as much as the median. The mean overhead is 14.22 % against a median of 7.90 %, and the interquartile range runs from 0 % to 20.6 %: a quarter of commutes need no detour whatsoever, while a tail pays substantially. Figure 4 shows one such case.

5.3 Geographic variation

State-level variation is extreme. Georgia commuters pass a median of eight cameras and pay 25.8 % overhead to avoid them; Pennsylvania commuters pass a median of one and pay 1.3 %, with 60 % passing any camera at all—up from 41 % two months earlier (Section 5.7).

A robust secondary finding: **avoidance is cheapest where cameras are densest.** Dense metropolitan grids offer a parallel street one block over, so the detour is short; rural and intercity routes often have one viable road, so avoidance means a genuine detour. The burden of refusal therefore falls hardest on drivers who have the fewest cameras to refuse—the reverse of the natural guess.

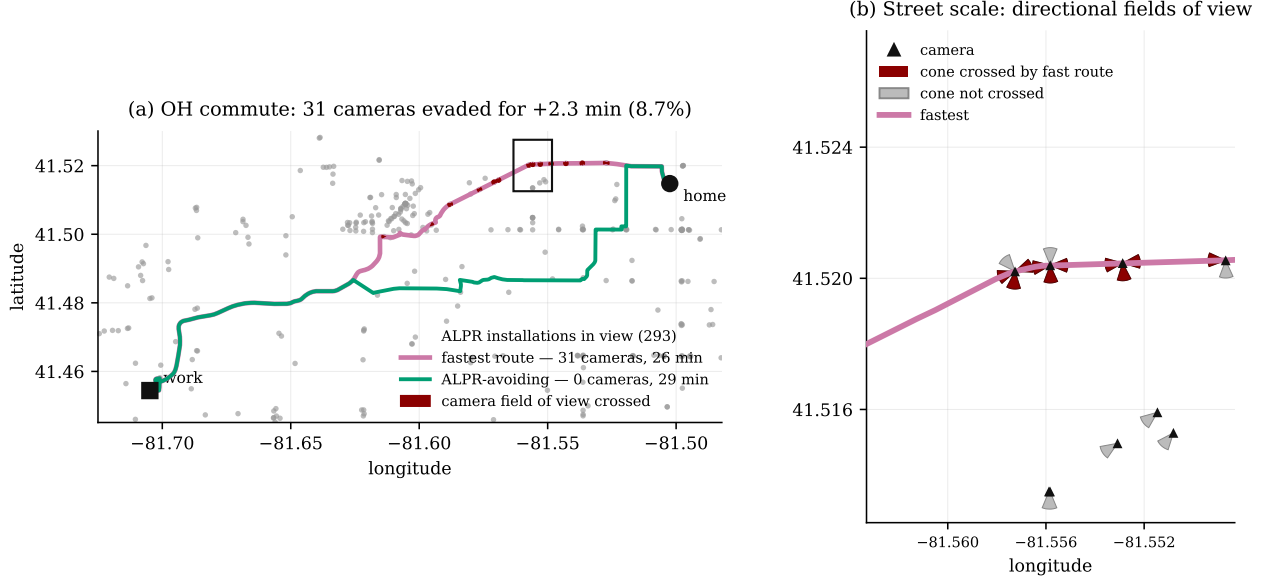


Figure 4: An illustrative high-exposure commute in the Cleveland area—chosen for legibility, not typicality. (a) The fastest route crosses 31 camera fields of view; the avoiding route crosses none, for 2.3 min (8.7%). (b) At street scale the directional model is visible: cones the fast route crosses are dark, cones aimed elsewhere are grey and cost nothing.

Table 4: National quartile contrasts. “Disjoint” marks non-overlapping 95% bootstrap CIs.

Contrast	Ratio	Cameras/km (Q1 vs Q4)	Verdict
Income Q1/Q4	1.00	0.093, 3 vs 0.124, 6	count overlaps; rate disjoint
% non-Hisp. Black Q4/Q1	1.36	0.084, 8 vs 0.139, 3	both disjoint
% Hispanic Q4/Q1	1.15	0.093, 3 vs 0.111, 6	both disjoint

5.4 Demographic contrasts

Both racial contrasts are statistically distinguishable nationally. Income is not, in counts—but higher-income origin tracts show materially *denser* per-kilometre exposure (0.124, 6 against 0.093, 3) offset by shorter commutes. Only reporting count and rate as peers surfaces that; a design treating the rate as a mere control would have discarded it.

5.5 Coverage-bias sensitivity, and why it governs the reading

Figure 7 is the most consequential figure in the paper. Recomputing quartiles *within each county* and pooling—so tracts are ranked only against neighbours subject to the same local mapping effort and broadly the same urban form—collapses the racial gradient from $1.36\times$ (disjoint CIs) to $1.05\times$ (indistinguishable), and leaves income at $0.92\times$ (indistinguishable). Within-county per-kilometre rates are close: 0.131, 1 against 0.142, 3.

The demographic gradients are therefore **entirely a between-county phenomenon**.

Mapped camera density varies enormously across the states studied—from roughly 108 per 100,000 residents in Georgia to 32 in Washington, a **$3.4\times$ spread** ($3.9\times$ in July, before Pennsylvania’s mapped density rose by half)—and state mean exposure tracks it closely (Figure 5b). From

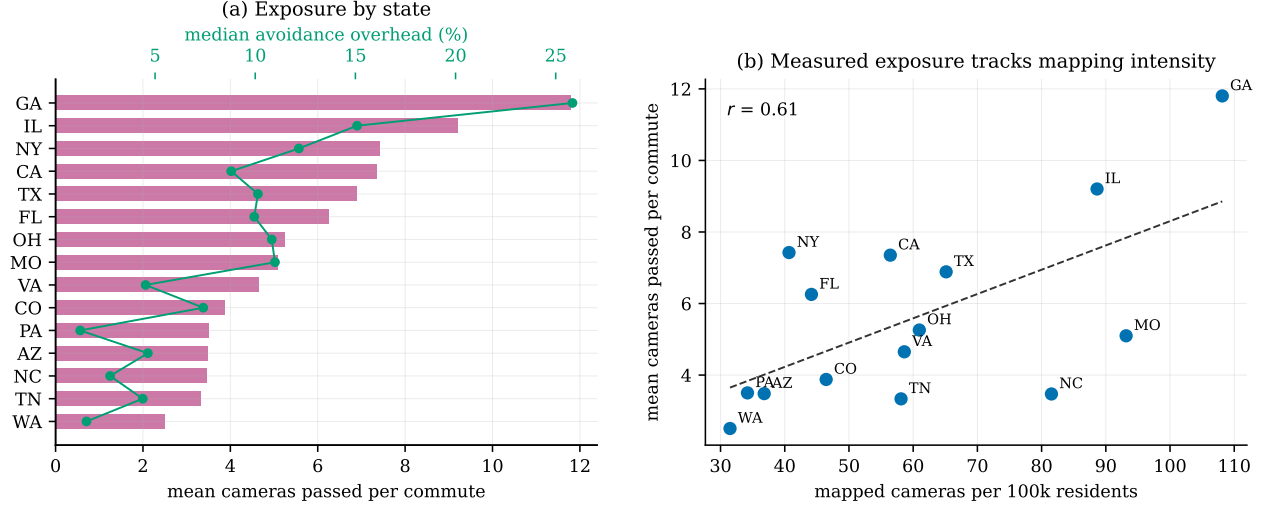


Figure 5: (a) Mean exposure by state with median avoidance overhead overlaid. (b) State exposure against mapped cameras per 100,000 residents. The relationship is close to linear, which is the central interpretive problem of Section 5.5: we cannot separate “more cameras” from “more mappers”.

Table 5: Vendor-resolved exposure over all 17,580 routes (98,898 camera encounters). Installation shares are of the 140,043 cameras that contribute cones, so they differ slightly from the 142,991-node denominator of Table 1. Flock holds four-fifths of installations and supplies two-thirds of exposure; per unit deployed, non-Flock cameras reach $2.3\times$ as many commuters.

Set	Installations	Exposures	Exposures/1,000 cameras	Routes ≥ 1	Evaded
Flock Safety	81.1 %	66.5 %	585	68.7 %	94.8 %
Other vendor	14.6 %	27.7 %	1,356	33.9 %	93.2 %
No vendor tag	4.3 %	5.8 %	964	12.6 %	96.9 %

one snapshot we cannot distinguish “Pennsylvania has few cameras” from “Pennsylvania has few mappers”; Section 5.7 shows what two snapshots can say.

Three explanations remain consistent with these data, and this design separates none of them:

1. **Differential mapping effort.** Volunteer contributors are unevenly distributed, and where they are absent cameras are absent from the map but present on the pole.
2. **Urban form.** High-%Black tracts are disproportionately urban and urban roads carry more cameras per kilometre, so the national contrast is substantially an urban/rural contrast.
3. **Genuine between-jurisdiction procurement differences.** Municipalities really do differ in what they buy.

We therefore decline to report a national demographic finding as a finding about surveillance siting. What we report is narrower and defensible: a gradient exists in the observable data, it is between-county, and it does not survive local ranking.

5.6 By camera vendor

Flock Safety holds 81.1 % of mapped installations but supplies only 66.5 % of measured commute exposure. Per unit deployed, non-Flock cameras generate **$2.3\times$** more exposure (Table 5). The

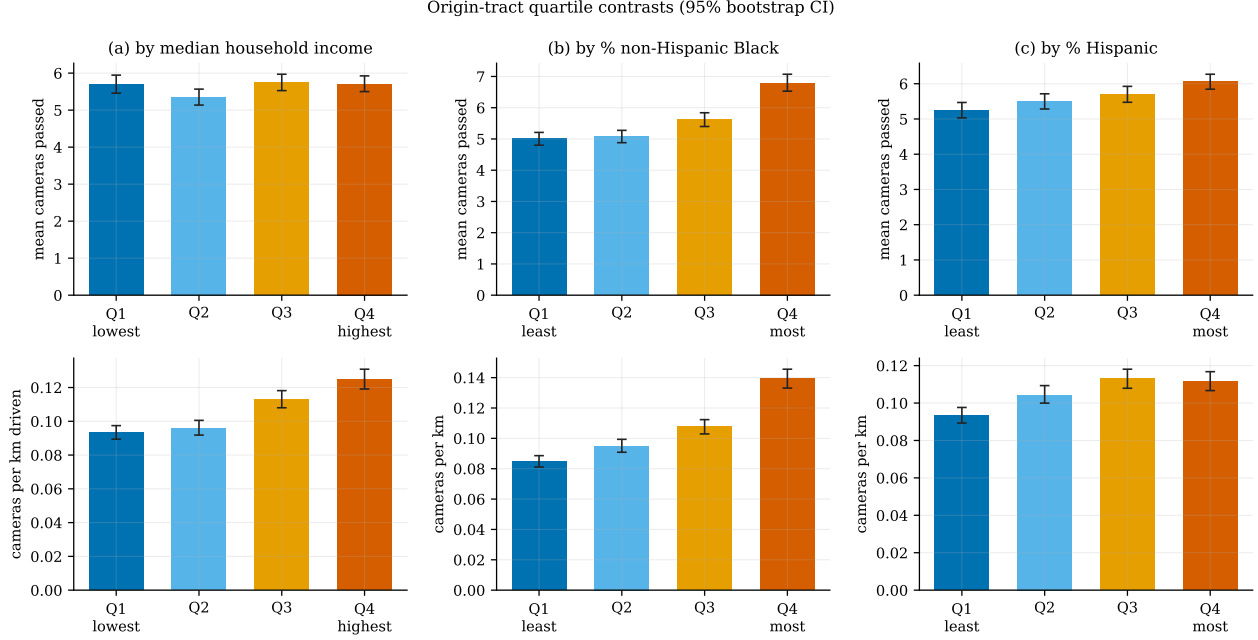


Figure 6: Origin-tract quartile contrasts, 95 % bootstrap CIs. Top row: mean cameras passed. Bottom row: cameras per kilometre driven. Income shows a per-km gradient with no count gradient—higher-income origins have denser but shorter commutes.

mechanism is visible in the cone geometry of Table 2: non-Flock units are multi-head more than six times as often, and DOT and tolling installations sit on the limited-access corridors commuters actually use, while a Flock unit watches one residential approach that fewer commutes cross.

This reframes the vendor’s dominance. Flock wins on count and loses on reach: its deployment model puts many cameras where comparatively few commuters pass, consistent with a product sold to municipalities and homeowners’ associations for neighbourhood coverage rather than to transport agencies for corridor coverage.

The gradient is stronger in transport infrastructure

Splitting Section 5.4 by vendor locates where the national disparity lives.

Table 6: Racial composition contrast (% non-Hispanic Black, Q4/Q1) resolved by vendor and then subjected to the within-county control of Section 5.5.

Camera set	National	Within-county	National verdict
Flock Safety	1.28	0.98	CIs disjoint
Other vendor	1.62	1.20	CIs disjoint, both
No vendor tag	1.07	1.03	indistinguishable
All vendors	1.36	1.05	CIs disjoint

Both major camera populations show a statistically distinguishable national gradient, but they are not equal: transport-agency and tolling cameras reach commuters from the highest quartile of Black population at $1.62\times$ the rate of the lowest, against $1.28\times$ for Flock. The disparity is roughly a quarter larger in infrastructure that no municipal council votes on.

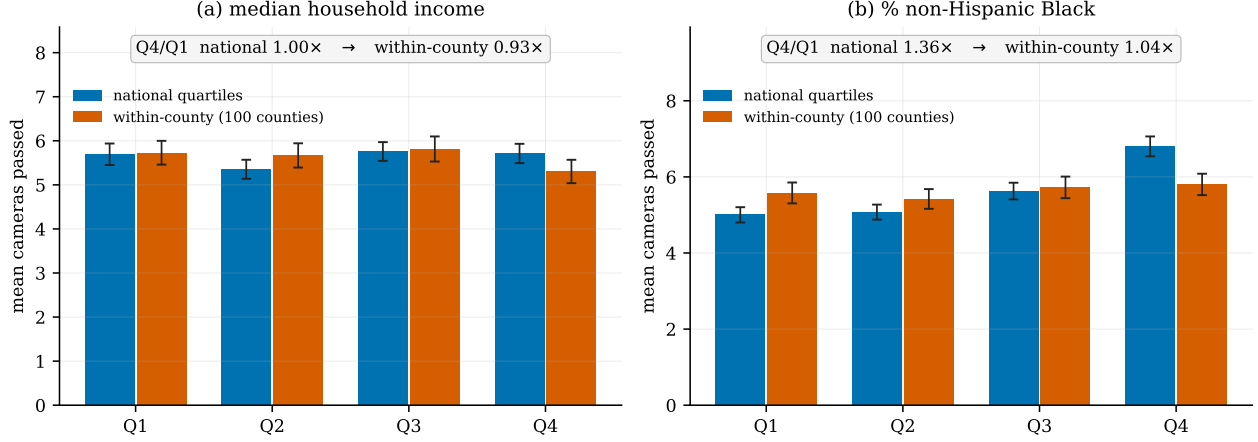


Figure 7: The central negative result. Ranking origin tracts within their own county rather than nationally collapses both demographic gradients to indistinguishability. 100 counties with $n \geq 40$ contribute 60 % of the sample.

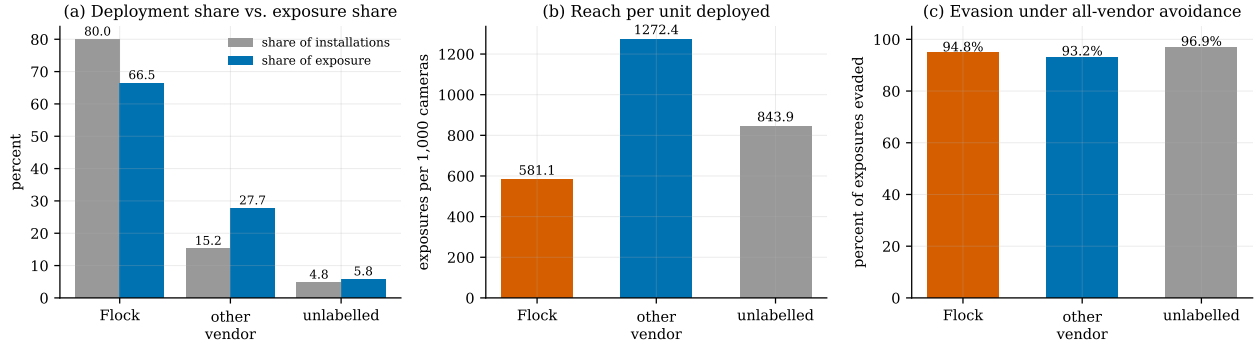


Figure 8: Vendor-resolved exposure. (a) Flock’s share of exposure falls well below its share of installations. (b) Per unit deployed, non-Flock cameras reach far more commuters. (c) Both are avoided at comparable rates under the all-vendor model.

A historical reading suggests itself. Mid-century urban highway construction routed limited-access corridors through Black neighbourhoods on a scale that is extensively documented, and tolling and DOT camera infrastructure sits on exactly those corridors. On that account the stronger gradient would be a decades-old artefact of road building rather than a contemporary artefact of camera procurement. We note the hypothesis; this design cannot test it.

And the discipline of Section 5.5 applies here, with one exception the July snapshot did not show. Under within-county ranking Flock’s gradient vanishes ($0.98\times$) and so does the untagged set’s ($1.03\times$), both with overlapping intervals. The transport-agency gradient narrows from $1.62\times$ to $1.20\times$ —and stays distinguishable, with intervals of 1.52 – 1.79 against 1.82 – 2.15 . It is the only demographic contrast in this paper that survives the control, and it survives narrowly: two months earlier, on 18 % fewer mapped cameras, the same contrast stood at $1.18\times$ with overlapping intervals. We report it as what it is: a within-county gradient in tolling and DOT exposure that is small, present in the September data, and consistent with the corridor reading above, since a limited-access highway routed through a Black neighbourhood is an intra-county pattern in a way that a municipality’s purchasing decision is not. Holding RQ5 to a weaker evidentiary standard than RQ3 would have let a between-county artefact return wearing a vendor label; holding it to the same

Table 7: The same 17,580 commutes against the two snapshots. Encounters grew by exactly the growth in mapped cameras; the marginal price of evading one did not move.

	2026-07-24	2026-09-22
Mapped cameras / cone parts	120,838 / 114,172	142,991 / 135,210
Camera encounters, all routes	83,565	98,898 (18.3 %)
Commutes passing ≥ 1 / ≥ 5 / ≥ 10	74.8 % / 34.6 % / 14.6 %	78.7 % / 39.7 % / 18.3 %
Cameras passed, median / mean	3 / 4.75	3 / 5.63
90 th / 99 th percentile	12 / 31	14 / 35
Exposure rate (cameras/km)	0.089	0.105
Reduced to zero	84.6 %	82.1 %
Extra time, median	1.50 min	1.98 min
Avoidance overhead, median	5.96 %	7.90 %
Minutes per camera evaded, median	0.78	0.78

standard is what lets this one through.

A correction, and what it cost. An earlier pass of this analysis ran on 15,043 of the 17,580 routes and reported that Flock exposure showed *no* racial gradient at all ($0.98\times$, overlapping; the July snapshot’s full-sample value was $1.20\times$, disjoint). That was wrong, and the error is instructive. The route-geometry sidecar is a gzip file appended to by more than one process, which makes it several concatenated gzip members; the reader used a single decompressor, which stops at the end of the first member and returns *successfully*. The missing 14 % was not merely noise—it moved Flock from “no gradient” to a disjoint gradient and would have supported a considerably stronger and more quotable claim than the data sustain. We record it because a silent truncation that returns success is the failure mode least likely to be caught by inspection, and because the corrected finding is the less convenient one.

5.7 Two months later: the same commutes against a fifth more cameras

Between the two snapshots—2026-07-24 and 2026-09-22—the mapped network grew from 120,838 to 142,991 nodes, 18.3 %. We re-ran the experiment against the September cones on the identical 17,904-commute sample and the identical road extract. The same 17,580 commutes routed (the same 324 did not), and every unavoided route was identical to the metre and the second, so the comparison is paired and everything that differs is camera data. Every other number in this paper is from the September run; this section is what changed.

Paired, the mean change is 0.87 cameras per commute (95 % CI 0.84–0.91). 35.5 % of commutes became more exposed, 59.7 % were unchanged, and 4.8 % became *less* exposed as cameras were retagged or removed; 830 commutes (4.7 %) that passed no camera in July pass at least one in September, and 133 went the other way. The zero-exposure set kept 14,251 commutes, lost 630 and gained 179. The paired change in avoidance cost is 0.95 minutes on average (CI 0.88–1.02), and a median of zero: most commutes pay exactly what they paid before.

Three things follow.

Measured exposure is linear in mapping effort. Encounters grew by 18.3 % against 18.3 % more mapped cameras. The new cameras were neither preferentially on commute routes nor preferentially off them, which is what one expects when the additions are contributors filling

in coverage rather than a vendor targeting corridors. It also says plainly what Section 7.2 argues: the absolute exposure figures are a lower bound that rises with the map, and will keep rising.

Refusal got dearer because there is more to refuse, not because each camera is harder to evade. The median cost rose by half a minute and the overhead by two points, while the median price per camera evaded stayed at 0.78 min to the second decimal. The road network’s capacity to route around a camera did not change; the count of cameras did. Four commuters in five still reach zero exposure for under two minutes.

The states that grew most are the ones we flagged as under-mapped. Pennsylvania’s mean exposure rose 88 %, its share of commutes passing any camera from 40.6 % to 60.2 %; Virginia rose 56 %, North Carolina 29 %, Texas 24 %. Georgia, Illinois, New York and California—already densely mapped in July—rose 9 % to 15 %. Two months earlier, Section 5.5 could not distinguish “Pennsylvania has few cameras” from “Pennsylvania has few mappers”. The map now answers, at least in part: the low-density states were the under-mapped ones, and a state’s July ranking said as much about its contributors as about its cameras. That is the within-dataset evidence for treating every cross-sectional demographic contrast in crowdsourced surveillance data as provisional.

6 Discussion

6.1 Refusal is cheap, and that is the policy-relevant fact

The strongest result here is also the simplest: for 82.1 % of American commuters, complete avoidance of licence-plate surveillance costs a median of 1.98 min. Arguments for ALPR deployment often rest implicitly on ubiquity—the idea that the network is comprehensive enough that avoidance is futile and therefore the privacy loss is unavoidable rather than chosen. At current mapped density that premise is false for four commuters in five—and two months of additional mapping moved that from five in six without touching the marginal price of a detour.

The corollary cuts the other way. If avoidance is this cheap, it is also cheap for people with reason to evade legitimate investigation, which bounds how much investigative value a fixed-camera network can deliver against a motivated subject while it continues to read every unmotivated one.

6.2 The equity question is not answerable from crowdsourced data

We set out to measure whether surveillance burden falls unevenly, and we can report only that the question is harder than it appears. The national gradient is real in the data, substantial ($1.36\times$), and statistically clean—and it evaporates under a control that any careful reviewer would demand, in every camera population but one, where a much smaller gradient remains.

We think this generalises. A considerable amount of accessible research on surveillance infrastructure necessarily rests on volunteer-mapped data, because no authoritative national inventory exists. Any such study that reports a demographic correlation without a within-locality control is at serious risk of reporting the demography of *mappers* rather than of the surveilled. We would encourage that control to become routine.

6.3 Attention and exposure are aimed at different infrastructure

Flock Safety is the name in municipal council debates, in local journalism, and in the activist mapping effort that produced the data underlying this paper. It holds four-fifths of mapped installations. Yet per unit deployed its cameras reach fewer than half as many commuters as a tolling or transport-agency camera, and the racial gradient in its exposure is roughly a quarter weaker.

We do not read this as exonerating, for a reason the measurement itself makes clear. A camera at the entrance of a residential subdivision does something qualitatively different from a camera on a toll gantry: it observes a small population repeatedly rather than a large population once, and repeated observation of a known small set is precisely what makes pattern-of-life inference possible. Commute exposure measures breadth, not depth, and Flock’s model appears optimised for depth. A study designed around repeat observation of the same vehicle would likely reverse the ordering reported here, and we would expect it to.

What the result does establish is that a distributional concern has been attached disproportionately to one vendor. The infrastructure with the steeper measured gradient is installed by transport agencies, largely absent from the municipal debate, and sited on corridors laid down decades before ALPR existed—and it is the one gradient that persists once tracts are compared only with their neighbours. Regulation aimed only at the vendor currently in the news would leave the larger measured disparity untouched.

6.4 Where the burden actually concentrates

The one distributional finding that does survive is geographic rather than demographic, and it is counter-intuitive: avoidance is cheapest in dense metropolitan grids and expensive in rural and intercity travel. Network topology, not camera count, sets the price of refusal. This is robust to the coverage critique in a way the demographic results are not, because it depends on road-network structure—which OSM maps close to completely—rather than on camera counts, which it does not.

7 Limitations

7.1 Tract centroids are not addresses

Every route runs centroid to centroid; real commutes begin in a driveway. In large or irregular tracts the centroid can sit far from any residence. This adds noise in both directions and is the main reason we report distributions rather than point estimates.

The stricter caution: tract demographics describe tracts, not the people routed. “Commutes originating in high-poverty tracts pass more cameras” is supportable; “poor people pass more cameras” is not.

7.2 Crowdsourced camera data is non-randomly missing

Absolute exposure figures are a **lower bound**: where contributors are absent, cameras are absent from the map but present on the pole. The hazard is not undercounting but its correlation with the variables of interest, quantified in Section 5.5. Every demographic statement in this paper is conditional on that, and the condition is not a formality.

7.3 Cone geometry: swept, and one result depends on it

The 60 m radius was chosen by reasoning—a pole set back from the kerb needs a cone long enough to reach the carriageway it watches—and the 45° default span from the 82% of explicit arcs that span exactly that. Neither was measured. We therefore swept the radius over {30, 45, 60, 90} m, rebuilding the cone geometry at each and re-scoring all 17,580 routes (Figure 9).

Two things follow, and they point in opposite directions.

Table 8: Cone-radius sensitivity. Exposure is mildly sensitive to the assumption; the avoidance result is not, until the radius grows by half.

Radius	Mean cameras	vs 60 m	Avoided routes still clean	Residual
30 m	4.21	0.75×	85.5 %	5.6 %
45 m	5.09	0.90×	83.6 %	5.4 %
60 m	5.62	assumed	81.6 %	5.6 %
90 m	6.29	1.12×	52.4 %	21.0 %

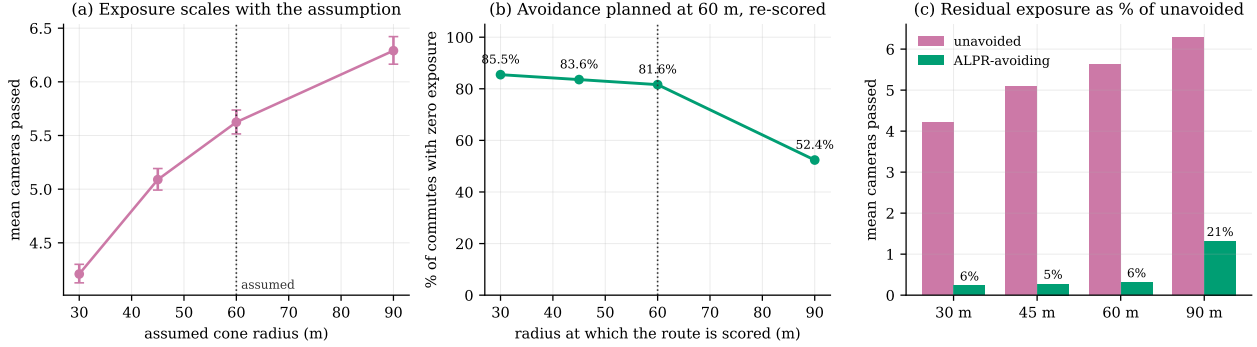


Figure 9: Sensitivity to the assumed cone radius. (a) Measured exposure scales gently—halving the radius removes only a quarter of it. (b) The share of commutes reaching zero exposure is flat below the assumed value and falls sharply above it. (c) Residual exposure as a fraction of unavoided, by radius.

Exposure is robust. Halving the radius removes only a quarter of the measured exposure, and 90 m adds 12 %. Cones are small relative to the road network at any of these values, so Section 5.1 does not rest on the choice.

The avoidance headline does depend on it. Routes planned against 60 m cones remain clean when re-scored at 45 m and 30 m—avoidance is not finely tuned to the assumption from below. But at 90 m the share reaching zero falls from 81.6 % to 52.4 %, a drop of 29.2 points (95 % CI $[-29.9, -28.6]$), and residual exposure quadruples from 5.6 % to 21.0 % of baseline. If real fields of view are half again as long as we assume, more than a third of the routes this paper counts as clean are not.

That figure is nonetheless a *lower* bound rather than an estimate, for a reason worth being precise about. These routes were *planned* against 60 m cones and only *re-scored* at 90 m; a router given 90 m cones would find different clean routes, and the denser the network the more of them there are. Measuring the true value requires a fresh graph import and a complete re-route at each radius—about nine hours apiece—which we have not done. What the sweep establishes is that the avoidance claim is conditional on the radius in a way the exposure claim is not, and that the conditionality bites only above the assumed value.

7.4 A single avoidance strength

All results use the maximum strength. Intermediate settings are the practically interesting ones for a shipped application and they behave badly: the usable range is roughly $0.3 \rightarrow 0.01$ and the dead zone is *density-dependent*. A setting of 0.3 does nothing in Delaware while avoiding 62.5 % of cameras in Atlanta. A single fixed slider-to-multiplier mapping is therefore not viable; the mapping must adapt to local density. This is an open problem for the application and an

unmeasured dimension of this paper.

7.5 Sample scope

Fifteen states, not fifty. Within-state commutes only (`od_main`), so cross-border metropolitan areas—New York/New Jersey, Philadelphia, Washington DC, Kansas City, St Louis, Portland—are systematically truncated at the state line. Several are exactly the dense multi-jurisdiction areas where the question is most pointed.

7.6 Two snapshots, two months apart

The network is expanding quickly—18 % more mapped nodes between 2026-07-24 and 2026-09-22—and absolute figures date accordingly; Section 5.7 shows measured exposure rising in proportion to the map. The relative claims held across the two snapshots: the cheapness of refusal (a marginal 0.78 min per camera in both), the metro/rural asymmetry, and the within-county collapse of the pooled gradient. Two months is a short baseline. The one conclusion that changed between them—the within-county transport-agency gradient becoming distinguishable—changed in the direction that growing coverage should move a real but small effect, and also in the direction it would move a mapping artefact concentrated in a few counties; a third snapshot would separate those. Routing uses free-flow speeds with no traffic, time-of-day, or turn-restriction penalties, so both travel times are optimistic. The *difference* between them—the quantity actually reported—is more robust than either absolute, since both share the bias.

7.7 Avoidance is a preference, not a guarantee

`multiply_by:` 0.01 makes observed roads expensive, not impassable. Where no alternative exists the router correctly uses the watched road. “Zero cameras” means the optimiser found a clean path, not that one was guaranteed to exist.

8 Reproducibility

Fixed seed 20260725 throughout, for both sampling and bootstrap. All external data sources are public and require no API key.

The 18 GB OSM extract and 14 GB graph cache are not redistributed; the acquisition and import scripts rebuild both.

Component	Path
Data acquisition	research/data/fetch.sh
Camera fetch (continental)	server/alpr/fetch-na-cones.sh
Cone construction	server/alpr/build_cones.py
Cone-set variants	server/alpr/build-variants.sh
Snapshot statistics	research/scripts/snapshot_stats.py
Routing configuration	server/graphhopper/config/trames.yml
Graph build	server/graphhopper/rebuild-with-cones.sh
Commute sampling	research/scripts/build_sample.py
Experiment	research/scripts/run_experiment.py
Geometry backfill	research/scripts/backfill_geometry.py
Vendor rescoring	research/scripts/vendor_exposure.py
Analysis	research/scripts/analyze.py
Post-processing, one command	research/scripts/run-analyses.sh
Snapshot comparison	research/scripts/compare_snapshots.py
Figures	research/paper/make_figures.py